INCIDENT LIFECYCLE AUTOMATION IN SERVICENOW

Project Documentation:

Project: Incident Lifecycle Automation in ServiceNow
Platform: ServiceNow Developer Instance
Module: Incident Management
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Project Role: ServiceNow Developer
Environment: ServiceNow Personal Developer Instance

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1. Project Overview

Incident Lifecycle Automation in ServiceNow is a ServiceNow-based IT Service Management project designed to demonstrate the complete lifecycle of an IT incident, from initial reporting and classification through investigation, escalation, resolution, and knowledge creation.

In many organizations, incidents may be reported through emails, phone calls, or informal communication channels. Without a structured Incident Management process, organizations may experience poor tracking, incorrect classification, delayed escalation, SLA breaches, and limited visibility into issue resolution.

This project demonstrates how ServiceNow can provide a centralized platform for managing incidents and connecting them with services, configuration items, knowledge articles, SLAs, and related incidents.

The project uses a Corporate VPN connectivity issue as the primary incident scenario. **Main**

Scenario

A user, **Michael Hoefer**, reports that he is unable to connect to the organization's Corporate VPN from his home office.

The incident is then:

Incident Creation

↓

Classification

↓

Knowledge Assistance

↓

Reassignment

↓

Level 2 Investigation

↓

Change Handling

↓

Child Incident Creation

↓

Resolution

↓

Knowledge Creation

↓

Final Validation

2. Phase 1 – Ideation

2.1 Problem Identification

The project was initiated to address the challenges associated with managing IT incidents without a standardized lifecycle.
Common problems include:

- Incidents being reported through multiple communication channels.
- Inconsistent incident classification.
- Incorrect assignment of incidents.
- Delayed escalation to specialized support teams.
- Difficulty tracking SLA commitments.
- Lack of reusable troubleshooting knowledge.
- Difficulty managing recurring issues affecting multiple users.
- Limited visibility into incident resolution history.

ServiceNow provides a centralized platform where incidents can be created, classified, assigned, investigated, resolved, and documented.

2.2 Proposed Solution

The proposed solution is to demonstrate an automated and standardized Incident Management lifecycle using ServiceNow.

The solution connects:

- Incident Management
- Service Management
- Configuration Management
- Knowledge Management
- SLA Management
- Related Incident Management
- Change Handling

2.3 Project Objectives

The major objectives are:

1. Enable Service Desk agents to create incidents.
2. Standardize incident classification.
3. Associate incidents with services and configuration items.
4. Provide knowledge assistance to support agents.
5. Reassign incidents to appropriate Level 2 teams.
6. Track incidents using SLA information.
7. Create child incidents for related user issues.
8. Document probable causes and resolutions.
9. Generate knowledge articles from resolved incidents.
10. Demonstrate an end-to-end Incident Management lifecycle.

3. Phase 2 – Requirement Analysis

3.1 Functional Requirements

The project requires the following functionality.

Incident Creation

The system must allow a Service Desk agent to create an incident containing:

- Caller
- Short description
- Description
- Assignment group
- Incident state

Incident Classification

The incident must support classification using:

- Category
- Subcategory
- Impact
- Urgency
- Service
- Service Offering
- Configuration Item

Knowledge Integration

Agents should be able to access relevant knowledge articles through Agent Assist and attach useful articles to the incident.

Incident Reassignment

The incident should be reassigned from the Service Desk to an appropriate Level 2 support group.

Level 2 Investigation

A Level 2 technician should be able to access:

- Incident information
- Knowledge articles
- SLA information
- Related records

Child Incident Management

The system should support creation of child incidents when another user experiences the same or related problem.

Incident Resolution

The incident should support:

- Probable cause
- Resolution code
- Resolution notes
- Resolved state

Knowledge Creation

A knowledge article should be created from the resolved incident.

SLA Tracking

The incident should maintain related SLA records that can be monitored throughout the lifecycle.

4. Stakeholder Analysis

Stakeholder	Role	Needs / Expectations	Project Benefit
End Users	Report IT issues	Quick resolution and visibility	Faster issue resolution
Service Desk Agents	First-level support	Easy logging and classification	Standardized incident intake
Level 2 Support	Technical investigation	Complete incident information	Better diagnosis
Change Management Team	Manage changes	Structured change handling	Controlled change process
ServiceNow Administrator	Platform management	Maintain configuration	Scalable process management

5. Non-Functional Requirements

The solution should provide:

Usability

The incident lifecycle should be easy for support agents to follow.

Traceability

Incident activities, assignments, SLA information, related incidents, and resolutions should be traceable.

Maintainability

The solution should use standard ServiceNow functionality wherever possible.

Visibility

Support teams should be able to monitor incident state, ownership, SLAs, and related records.

Reusability

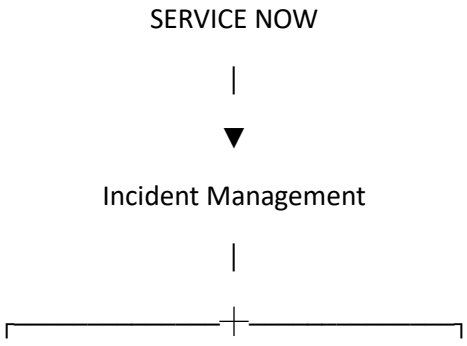
Resolved incidents should be convertible into knowledge that can help resolve future incidents.

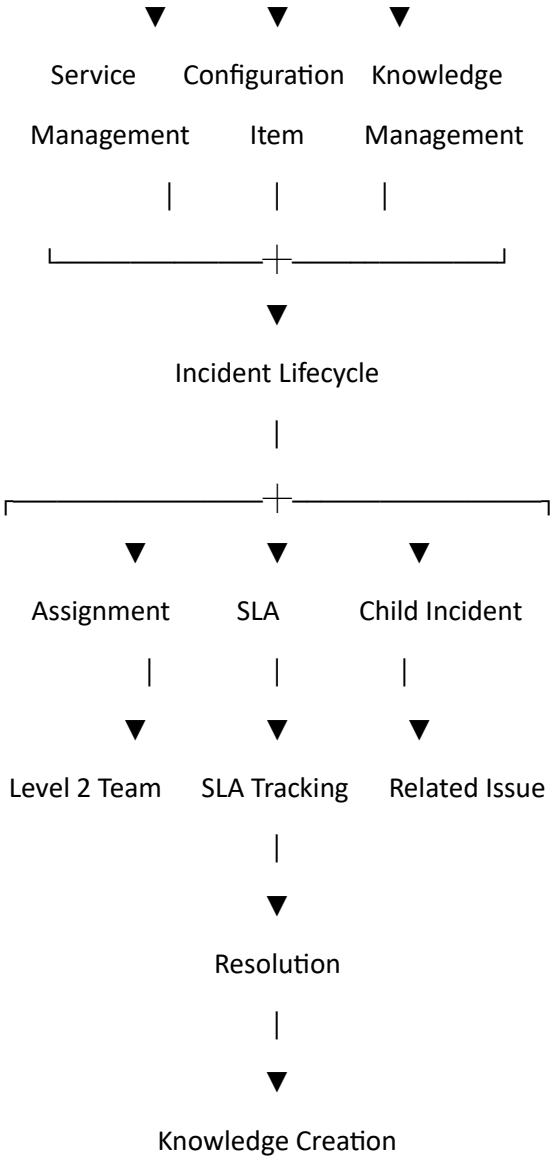
6. Phase 3 – Project Design

6.1 System Design

The project uses ServiceNow's existing Incident Management capabilities and connects the incident to other ServiceNow records.

High-Level Architecture





6.2 Service Hierarchy

The project establishes the following service hierarchy:

Remote Access

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└— Corporate VPN

The configuration item associated with the incident is:

PowerEdge T110 II

This provides a relationship between the incident, service offering, and affected configuration item.

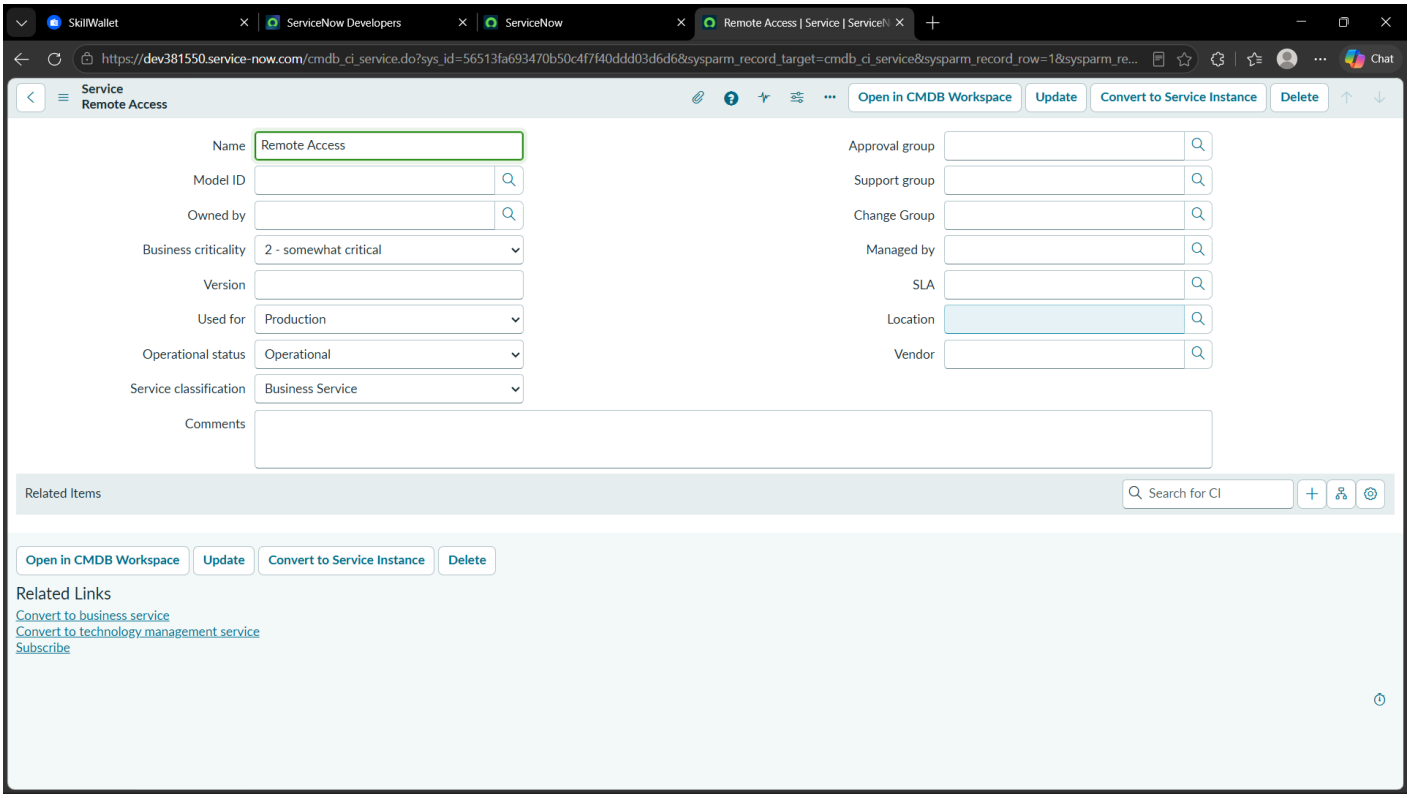


Figure 1. Service configuration for the Remote Access service in ServiceNow.

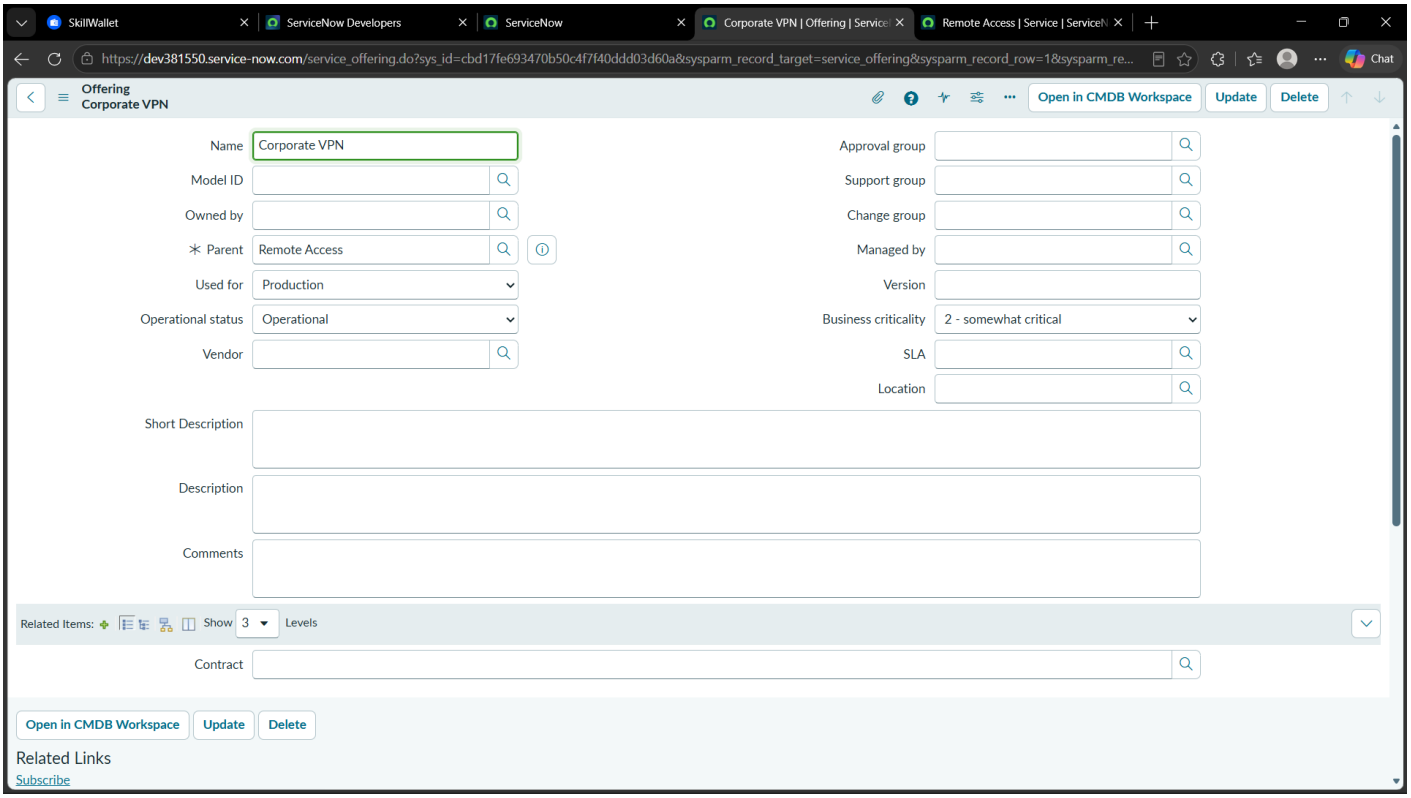


Figure 2. Corporate VPN service offering associated with the Remote Access service.

7. Incident Lifecycle Design

The designed incident lifecycle is:

New

↓

In Progress

↓

Classified

↓

Knowledge Assisted

↓

Reassigned to Network

↓

Level 2 Investigation

↓

On Hold – Awaiting Change

↓

Child Incident Tracking

↓

Resolved

↓

Knowledge Creation

↓

Final Validation

8. Phase 4 – Project Planning

8.1 Development Plan

The project was planned as a sequence of functional milestones.

Phase	Activity
1	Requirement analysis
2	Service creation
3	Service offering creation
4	Incident creation
5	Incident classification
6	Knowledge integration
7	Reassignment

8	Level 2 investigation
9	Change handling
10	Child incident creation
11	Incident resolution
12	Knowledge creation
13	Final validation
14	Documentation

8.2 Project Resources

Platform

ServiceNow Developer Instance

Main Module

Incident Management

Supporting Modules / Features

- Service Management
- Configuration Management
- Knowledge Management
- SLA Management
- Service Operations Workspace

8.3 Project Roles

Service Desk Agent Responsible

for:

- Incident creation
- Classification
- Initial investigation
- Reassignment

Level 2 Technician

Responsible for:

- Advanced investigation
- Incident ownership
- Resolution

End User

Responsible for reporting the issue.

ServiceNow Administrator

Responsible for maintaining the ServiceNow configuration.

9. Phase 5 – Project Development

9.1 Service Creation

A new service named **Remote Access** was created.

Configuration:

Field	Value
Name	Remote Access
Operational Status	Operational

9.2 Service Offering Creation

A service offering named **Corporate VPN** was created under the Remote Access service.

Field	Value
Name	Corporate VPN
Parent	Remote Access
Operational Status	Operational

9.3 Incident Creation

The incident was created from the **Service Operations Workspace**.

Incident Details

Field	Value
Incident Number	INC0010004
Short Description	Unable to connect to Corporate VPN from home office
Caller	Michael Hoefer
Assignment Group	Service Desk

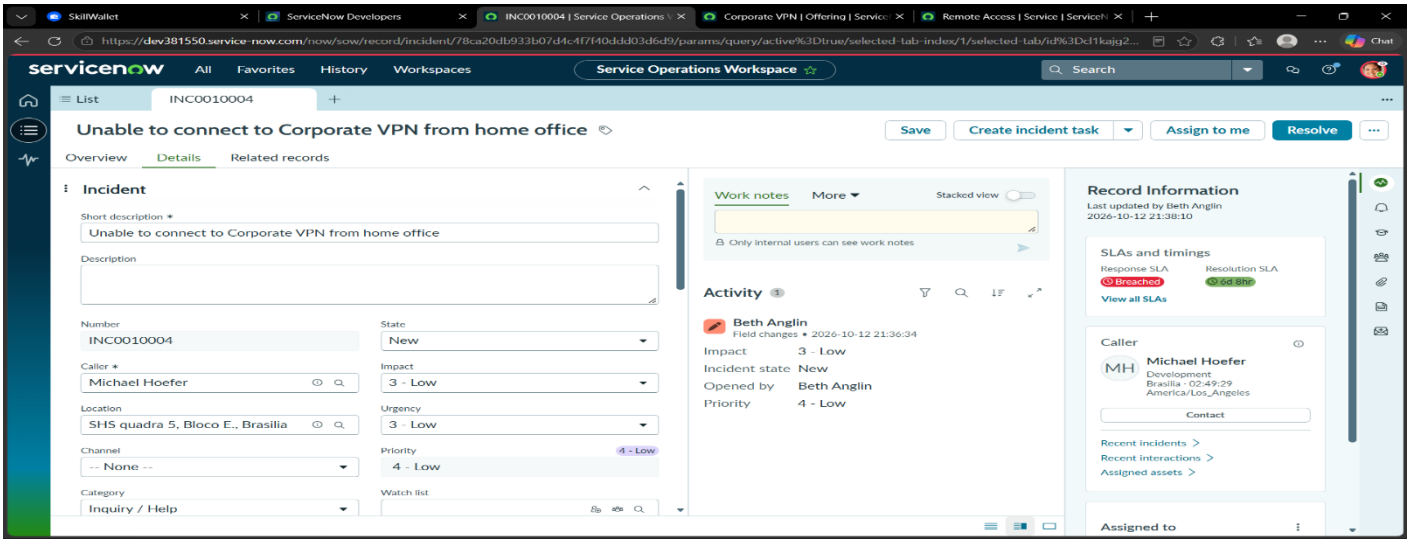


Figure 3. Incident INC0010004 created for the Corporate VPN connectivity issue.

10. Incident Classification

The incident was classified using the following values:

Field	Value
Category	Network
Subcategory	VPN
Urgency	2 - Medium
Service	Remote Access
Service Offering	Corporate VPN
Configuration Item	ThinkStation S20
Channel	Phone

The incident description was updated to: Michael advised he was able to connect yesterday evening, but this morning received an authentication error. The Internet works fine.

A work note was also added: Classified and assigned incident.

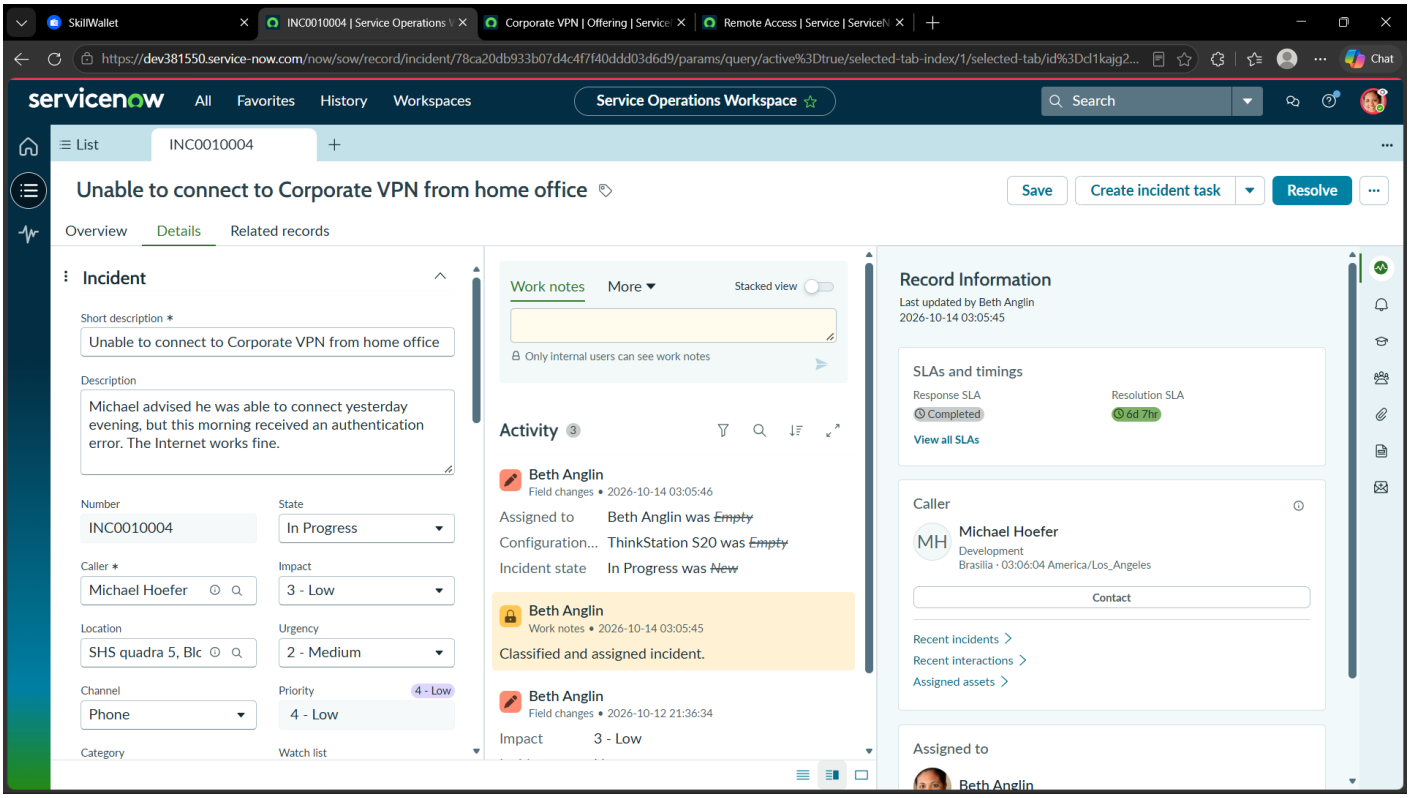


Figure 4. Incident classification with Network/VPN categorization, urgency, service, service offering, and configuration item.

11. Knowledge Integration

Agent Assist was used to identify relevant knowledge articles based on the incident description.

The knowledge article was opened and marked as **Helpful**.

The article was attached to the incident with the comment:

Follow the Steps and Resolve the Incident.

This demonstrates how existing organizational knowledge can assist Service Desk agents in troubleshooting incidents.

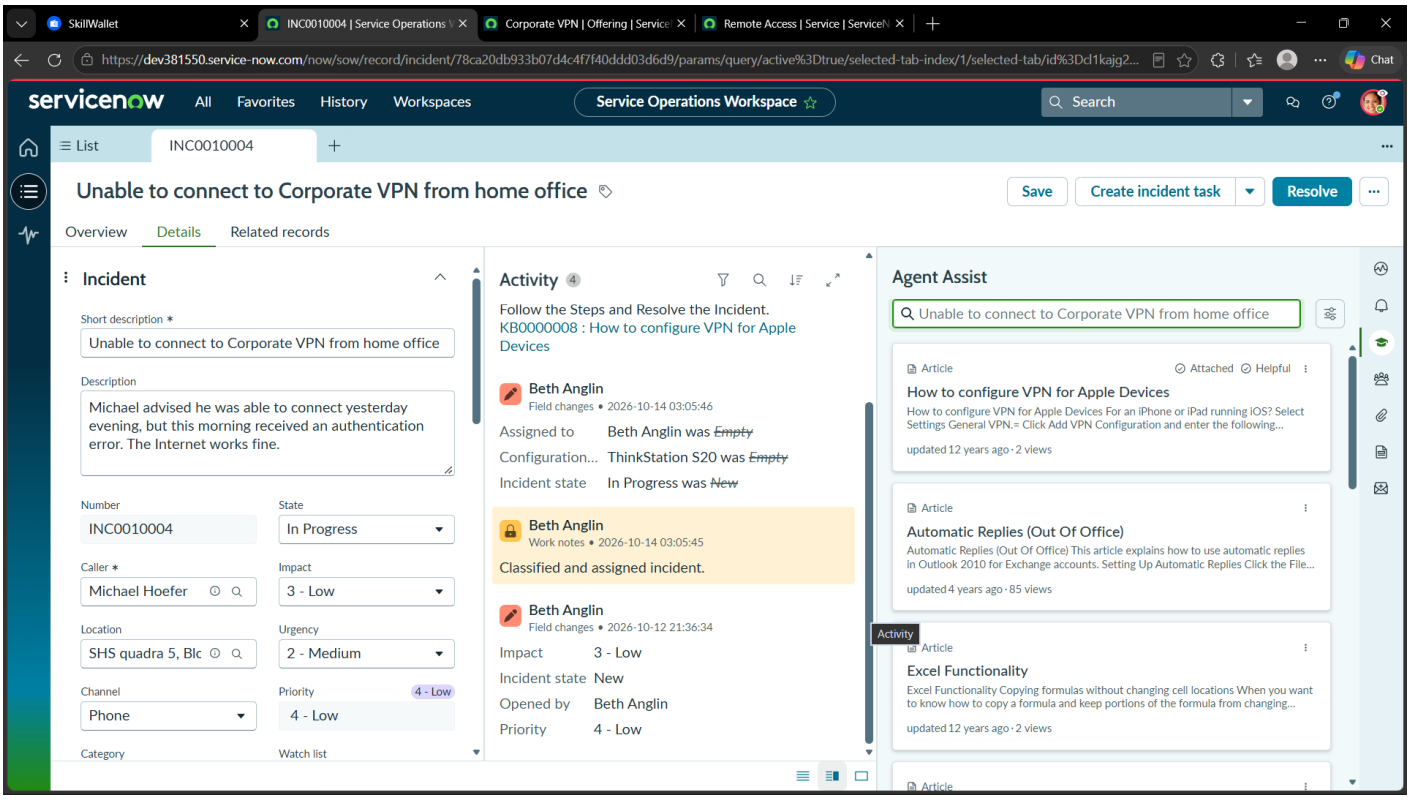


Figure 5. Agent Assist displaying relevant knowledge articles for the incident.

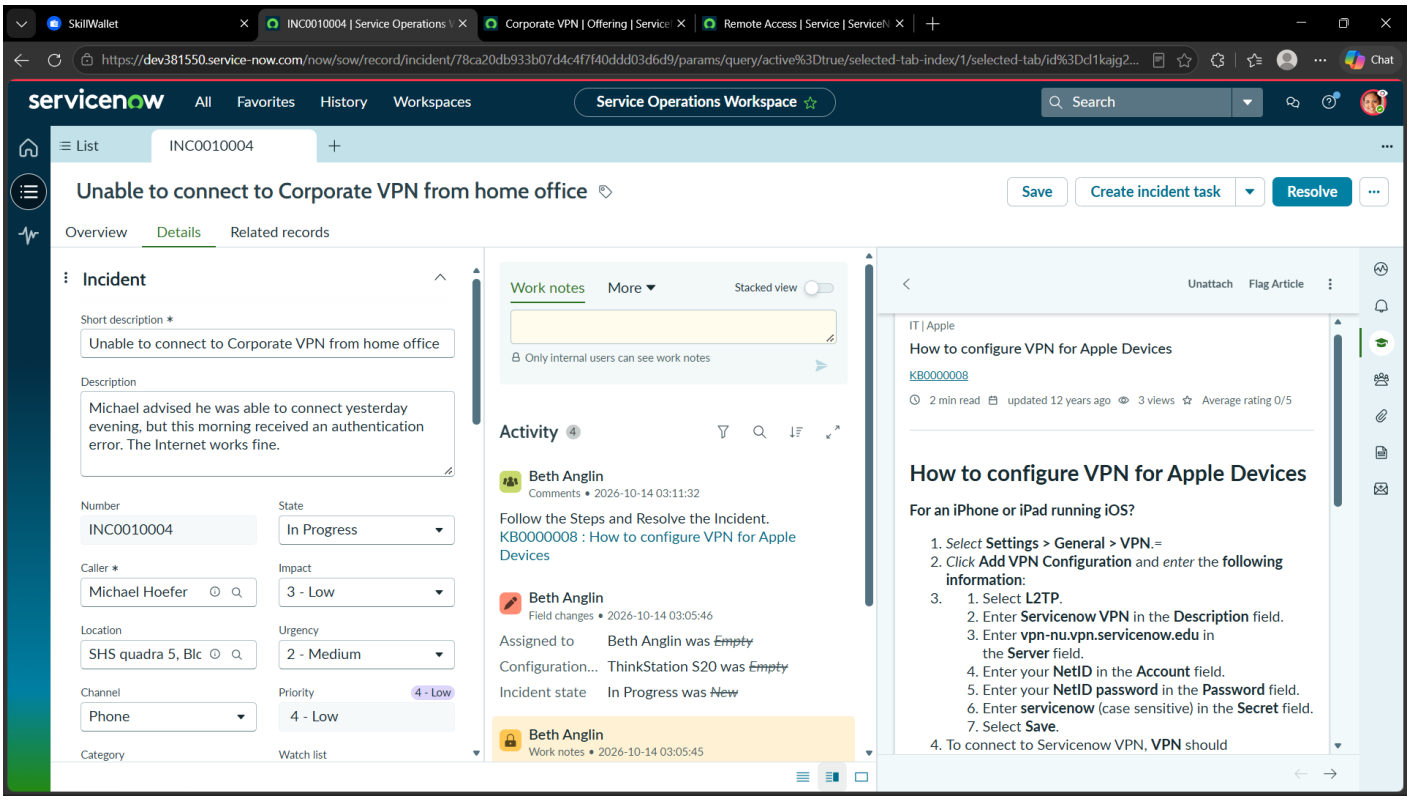


Figure 6. Knowledge article attached to the incident after being marked helpful.

12. Reassignment and Escalation

The incident was initially handled by the **Service Desk**.

The incident was then reassigned to:

Assignment Group: Network

The assignment was cleared automatically when the assignment group changed.

A watch list member and work notes list member were also configured to allow appropriate users to monitor the incident.

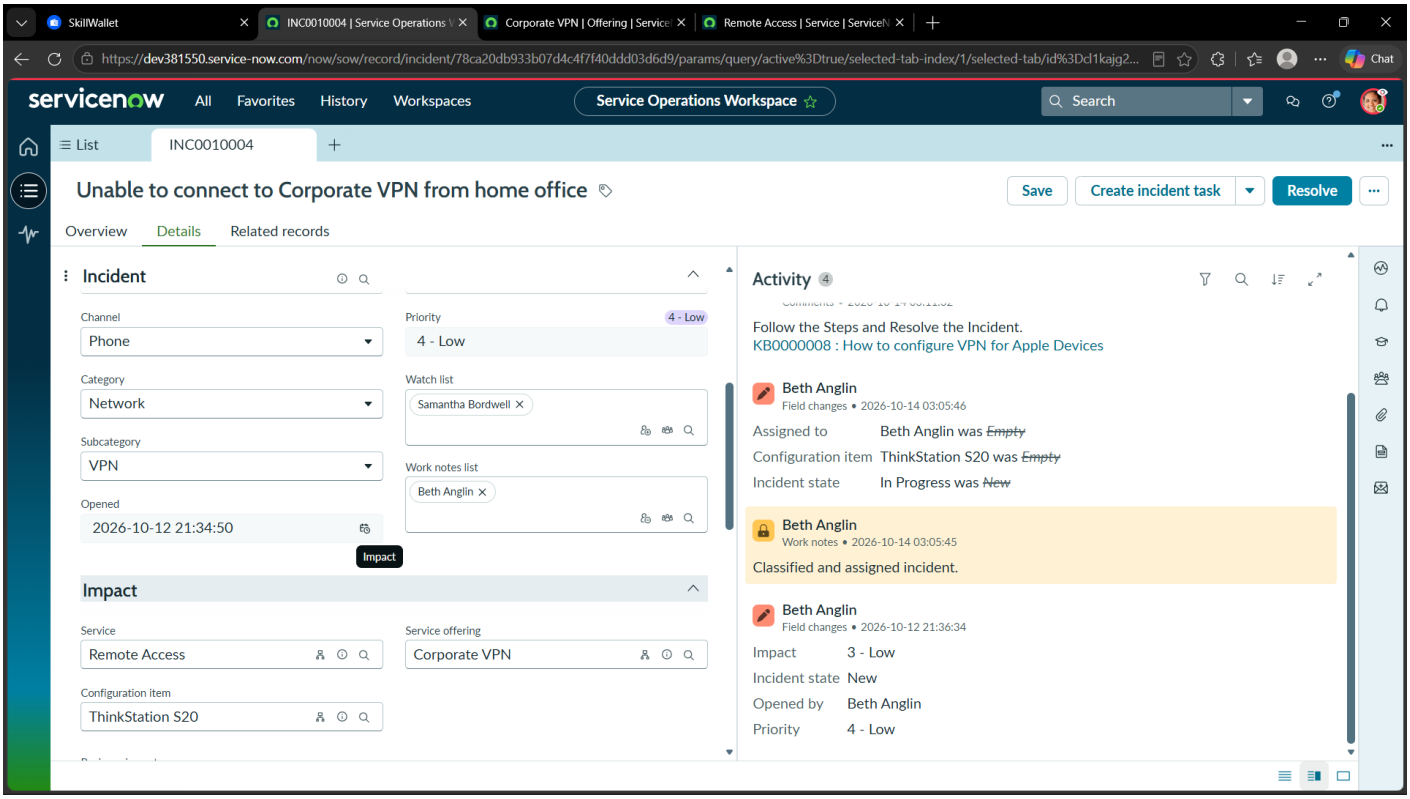


Figure 7. Incident collaboration users added to the Watch List and Work Notes List.

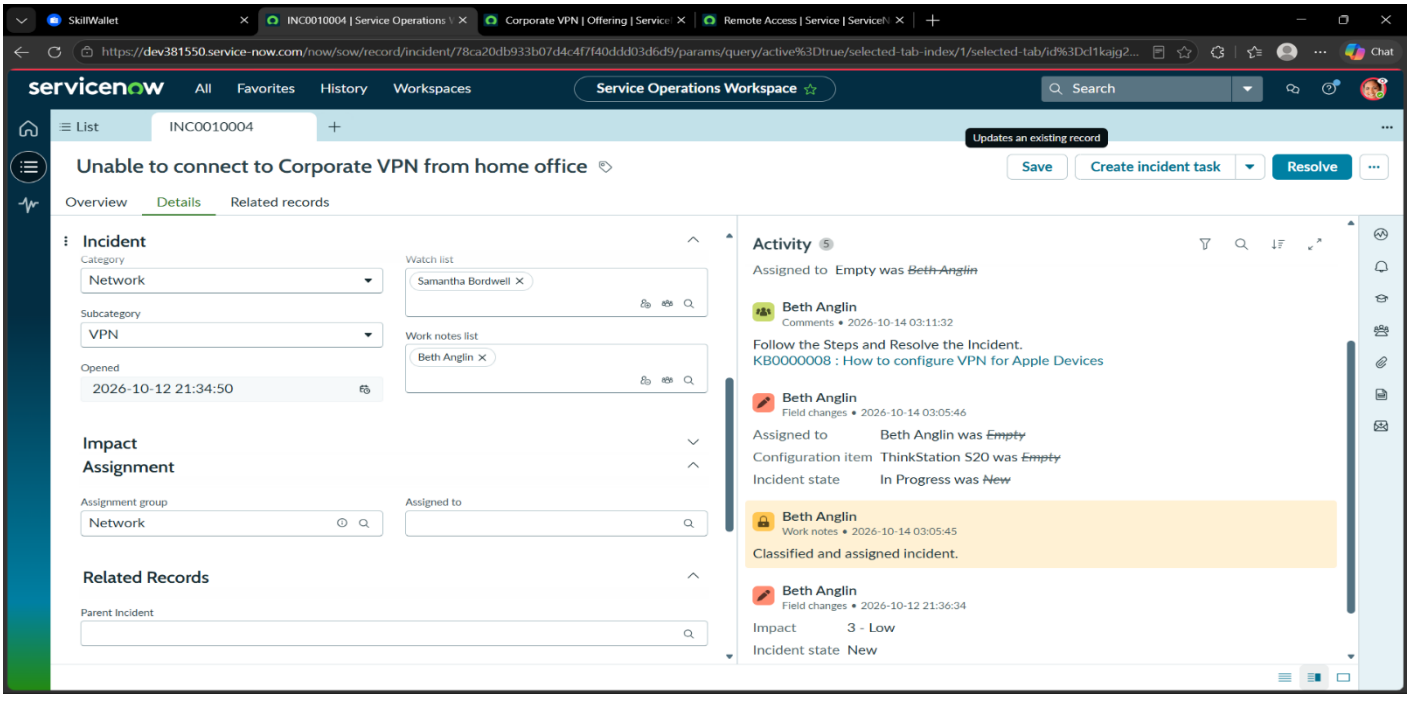


Figure 8. Incident reassigned from Service Desk to the Network support group.

13. Level 2 Investigation

The incident was then investigated by a Level 2 technician.

The user **David Loo** was used to demonstrate the Level 2 support workflow.

The technician was able to access:

- Incident description
- Knowledge information
- SLA information
- Related records

The incident was assigned to David Loo for investigation.

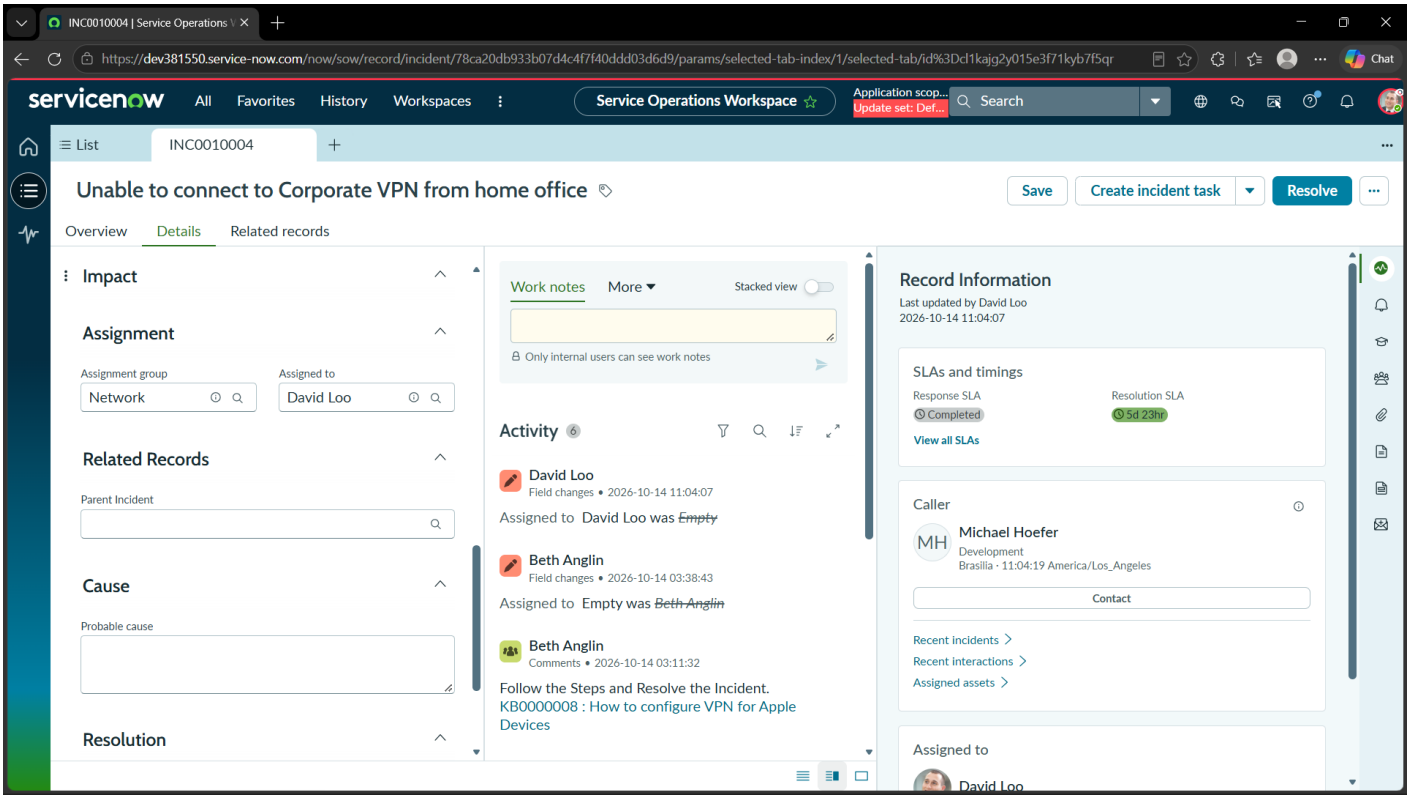


Figure 9. Level 2 technician David Loo investigating the incident.

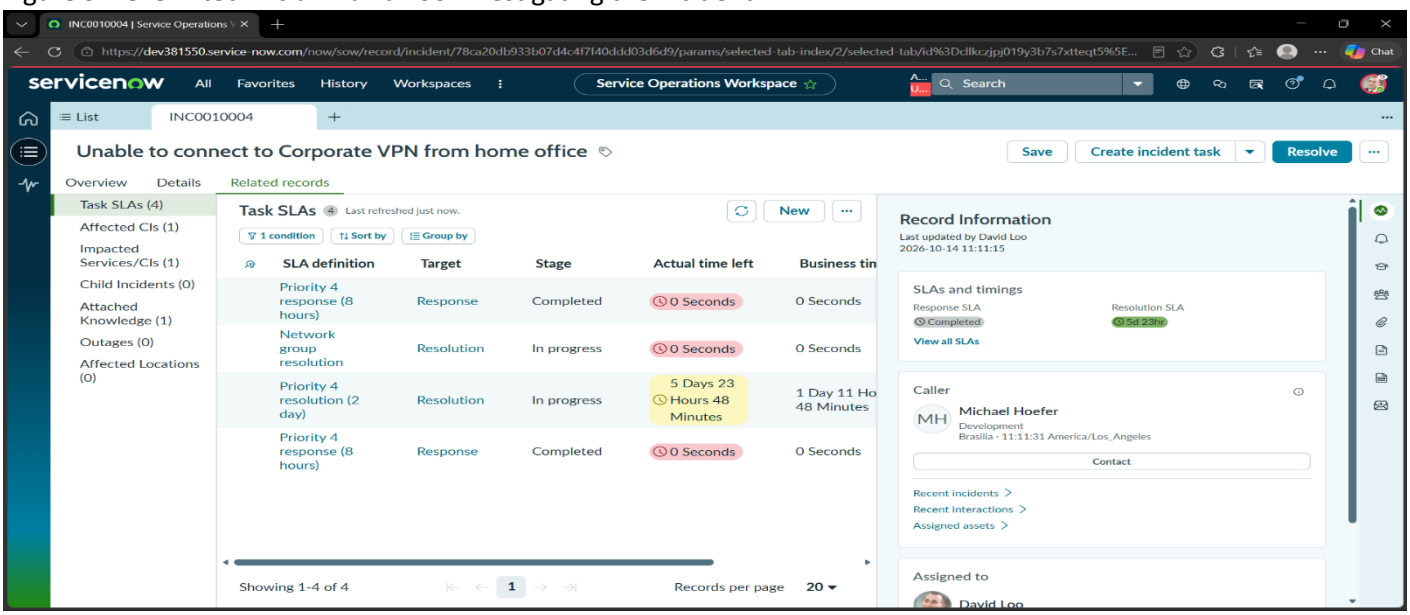


Figure 10. SLA information associated with the incident and visible during Level 2 investigation.

14. Change Handling

During investigation, the affected configuration item was updated to: **PowerEdge**

The incident was then placed on hold with:

State: On Hold On Hold Reason: Awaiting Change

This demonstrates how an incident can temporarily pause resolution while required change-related work is pending.

Note: A separate Change Request was not created because the supplied project execution steps only required the incident to be placed on hold with the reason "Awaiting Change."

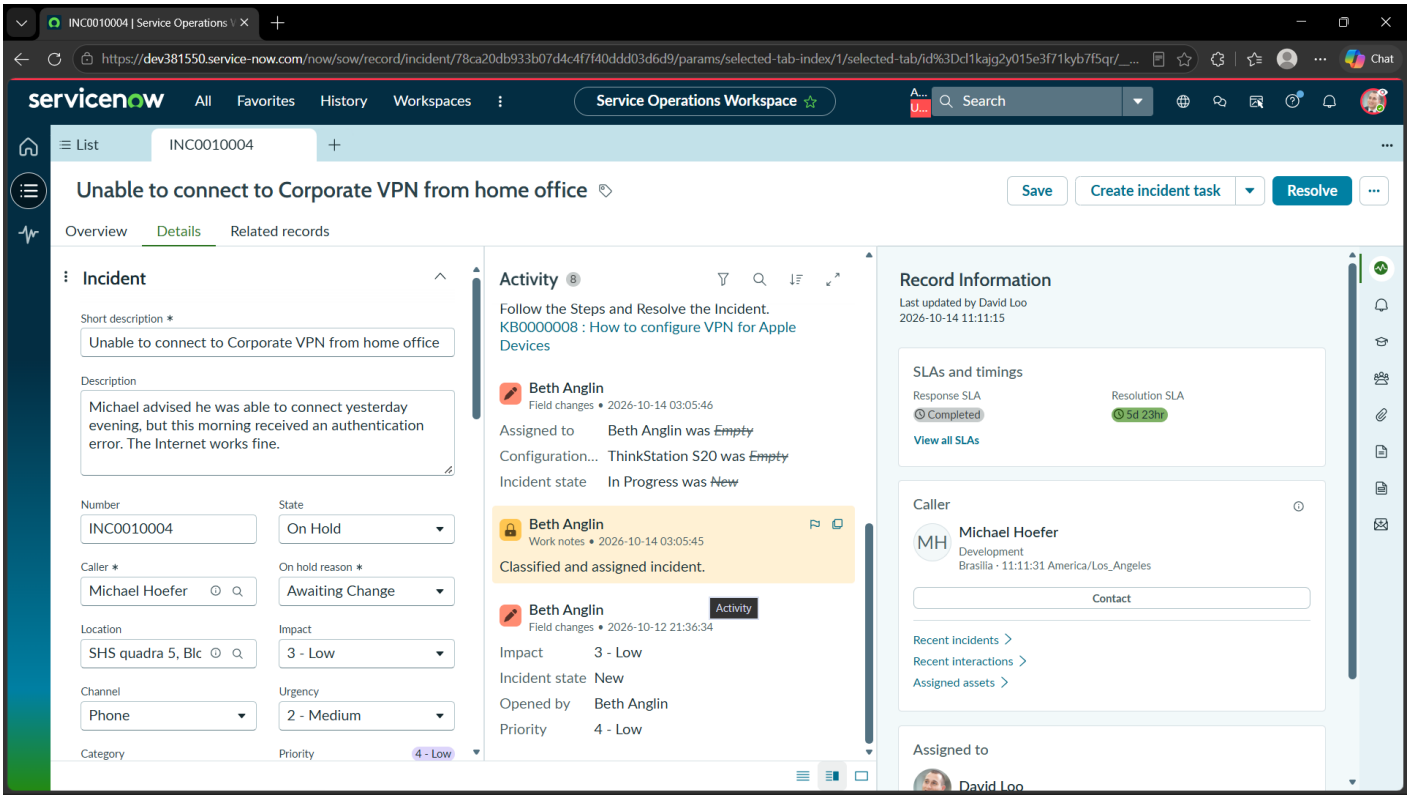


Figure 11. Incident placed in the On Hold state with the reason "Awaiting Change" after updating the affected configuration item.

15. Child Incident Creation

A child incident was created from the parent incident to demonstrate management of a related user issue.

Parent Incident INC0010004 Child Incident INC0010005 Child Incident Details

Field	**Value**
Short Description	Unable to connect to Corporate VPN
Caller	Abel Tuter
Description	User receiving VPN authentication failure error

The child incident was linked to the parent incident through the **Child Incidents** related list.

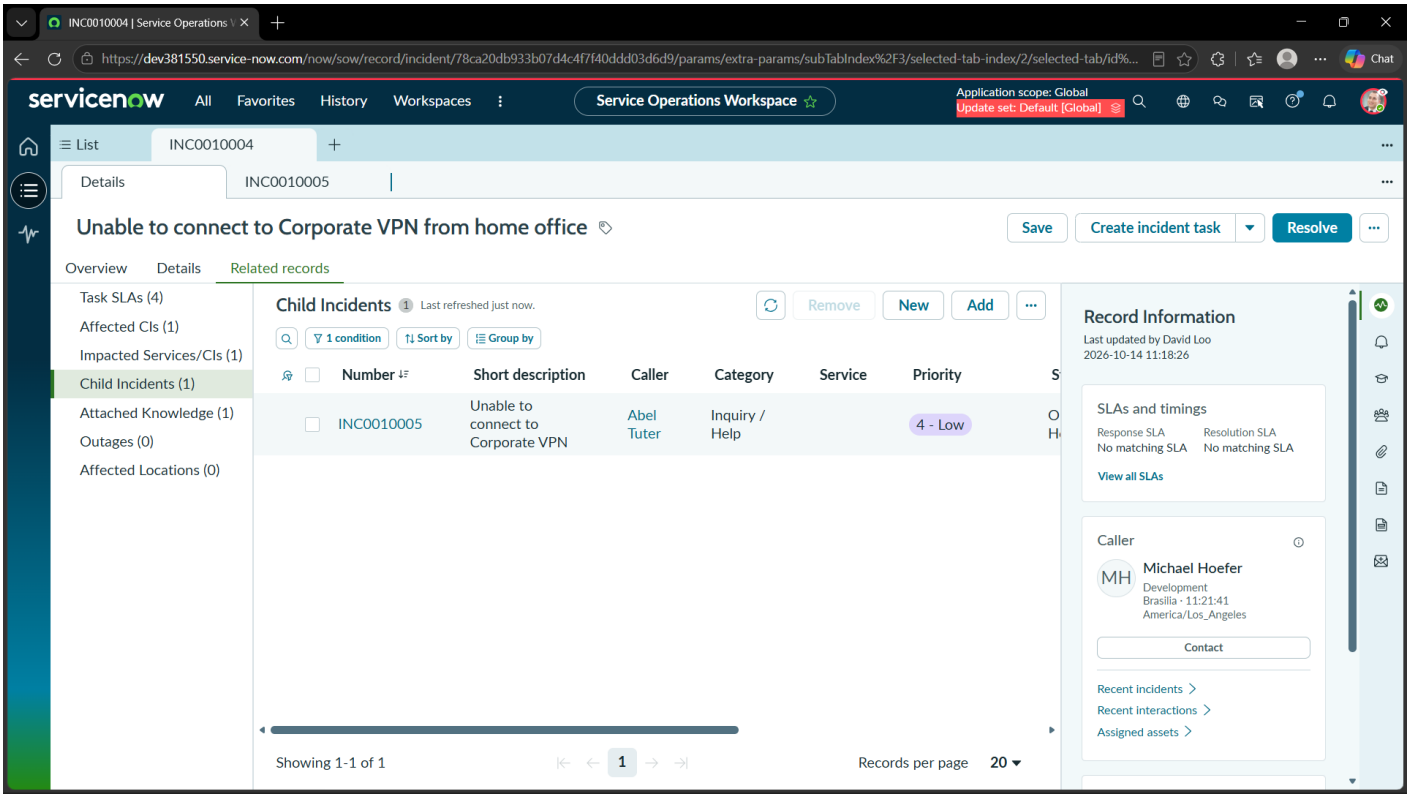


Figure 12. Child incident INC0010005 created from parent incident INC0010004.

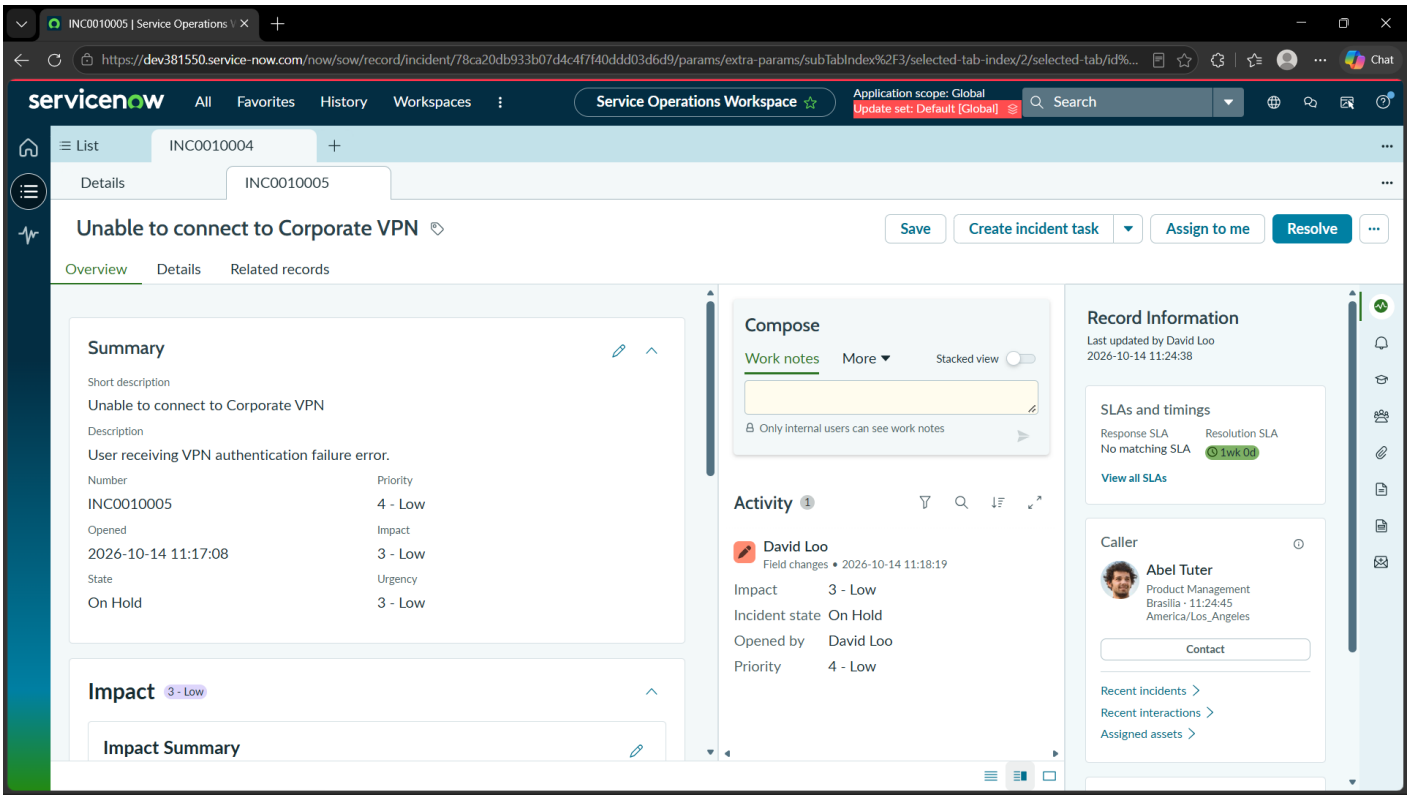


Figure 13. Child incident INC0010005 displayed under the Child Incidents related list of the parent incident.

16. Incident Resolution

The probable cause was documented as:

PowerEdge service was suspended and required restart.

The resolution code was:

Workaround provided

Resolution notes:
Restarted VPN-SRV-02 service as per emergency change request.

The incident was then moved to the **Resolved** state.

The activity history recorded the transition from **On Hold → Resolved**.

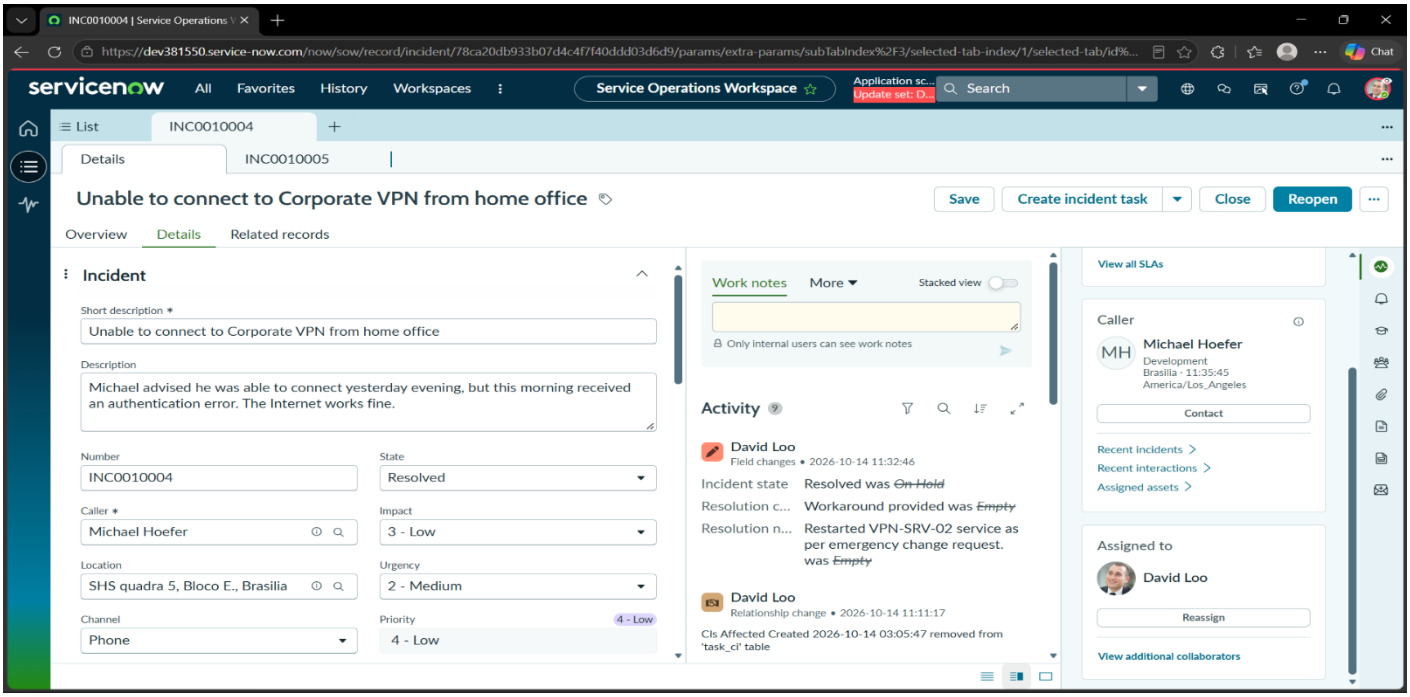


Figure 14. Parent incident INC0010004 resolved with documented cause and resolution information.

17. Child Incident Resolution

The child incident **INC0010005** was also verified as resolved.

This demonstrates that related incidents can be independently tracked through their own lifecycle while maintaining their relationship with the parent incident.

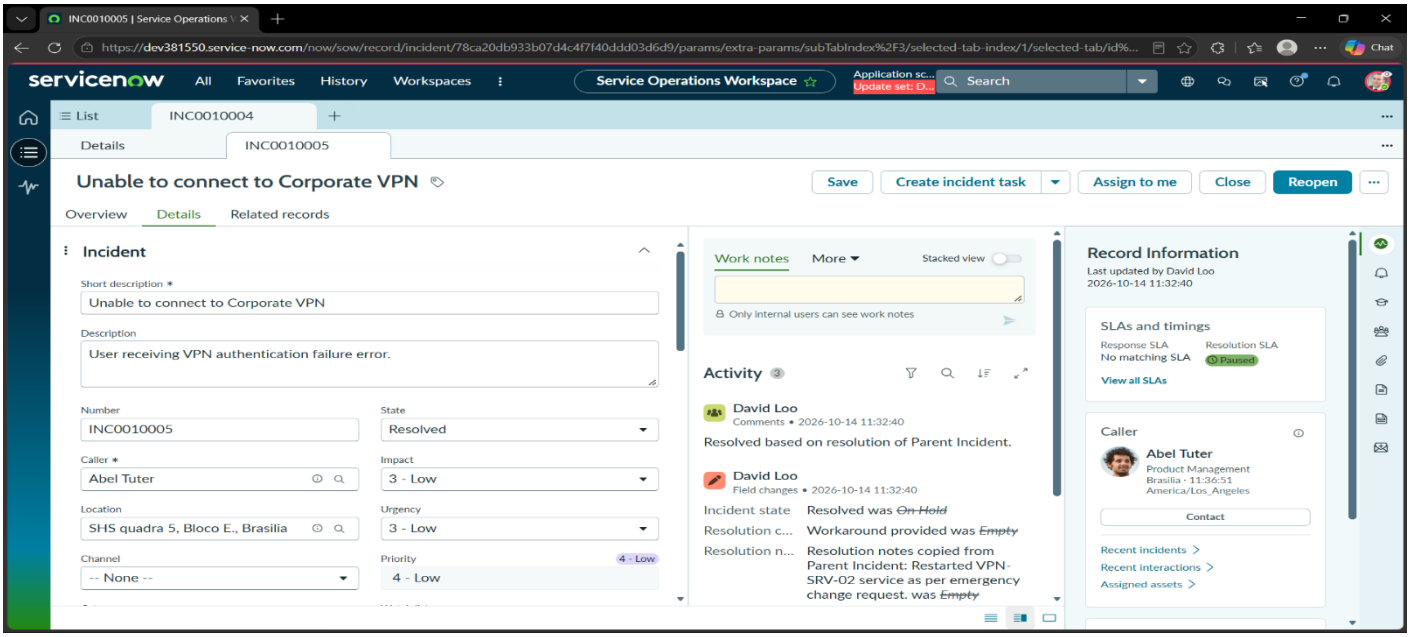


Figure 15. Child incident INC0010005 verified in the Resolved state.

18. Knowledge Creation

After resolving the incident, a new knowledge article was created from the incident.

Configuration Knowledge Base: IT

Template: Standard

The article was reviewed and saved.

The parent incident subsequently displayed:

- Attached Knowledge
- Created Knowledge
- Related incident
- Affected CI
- SLA information

This demonstrates how incident resolution knowledge can be converted into reusable organizational knowledge.

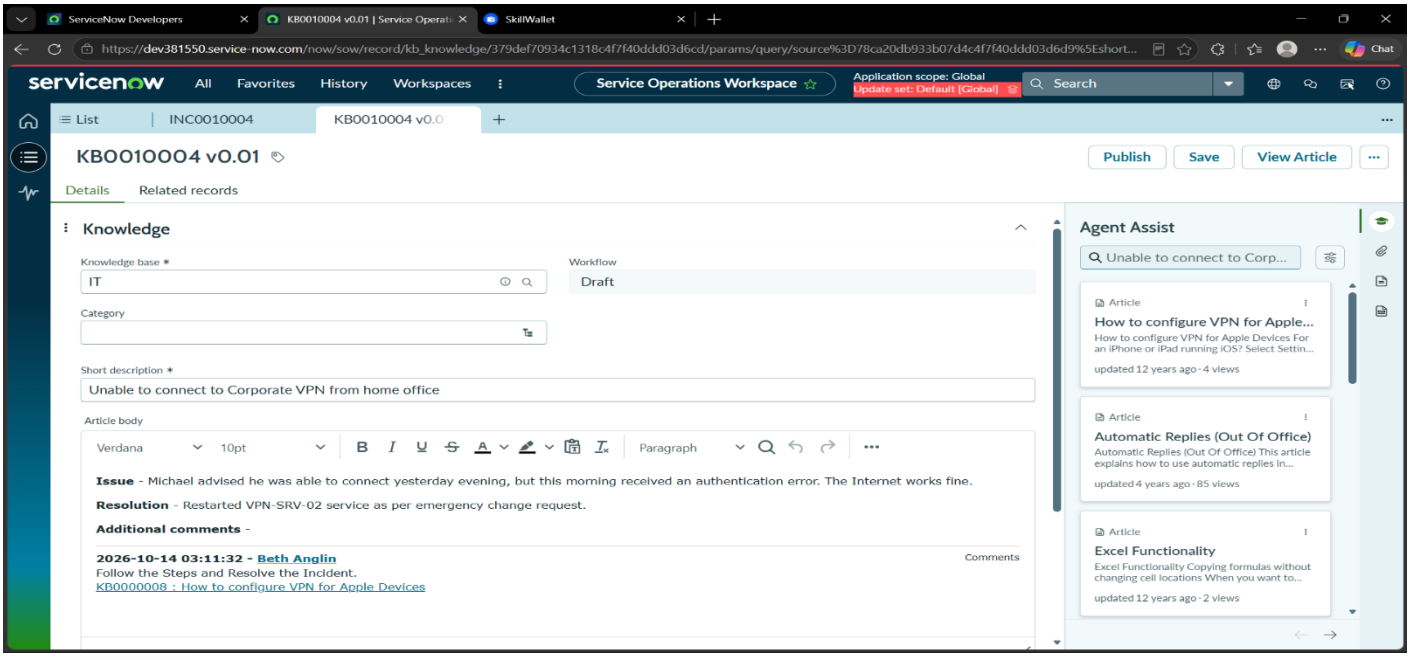


Figure 16. Knowledge article created from the resolved incident using the IT Knowledge Base and Standard template.

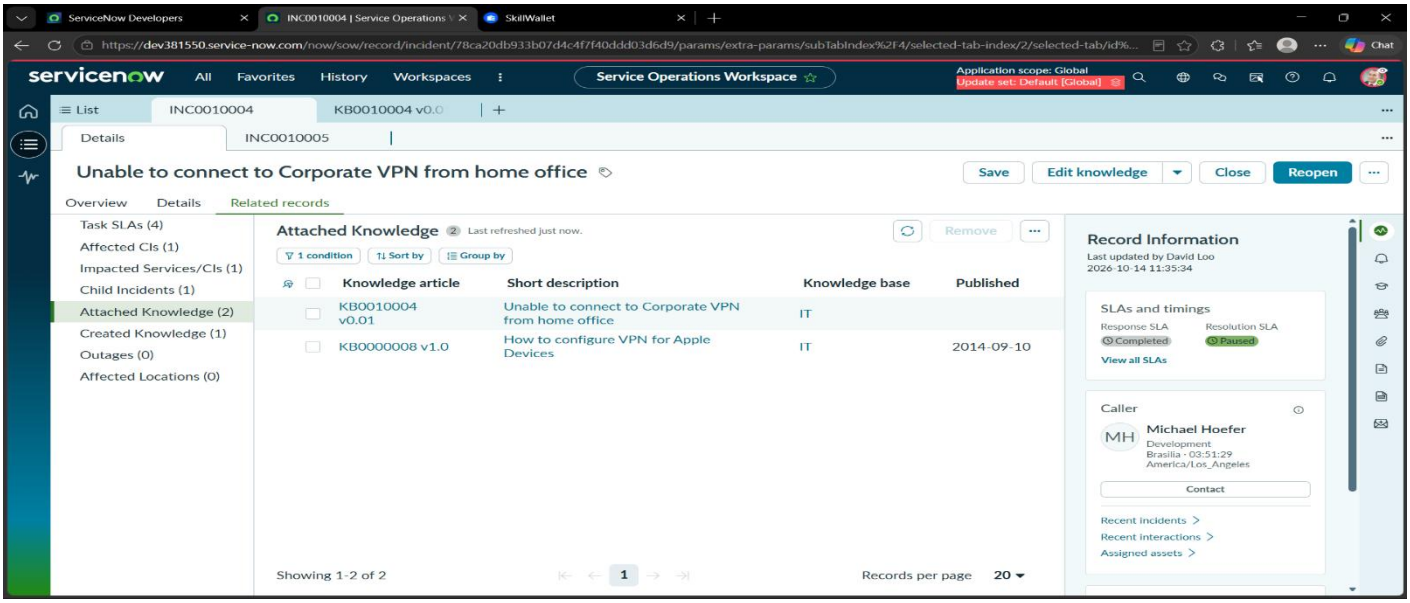


Figure 17. Created Knowledge record visible under the parent incident's Related Records section.

19. Final Incident Validation

The final parent incident was reviewed through the **Related Records** section.

The following records were visible:

- Task SLAs
- Affected CIs
- Impacted Services/CIs
- Child Incidents
- Attached Knowledge
- Created Knowledge

The affected configuration item was displayed as:

PowerEdge T110 II

The parent incident was assigned to:

David Loo

The parent incident was in the:

Resolved state.

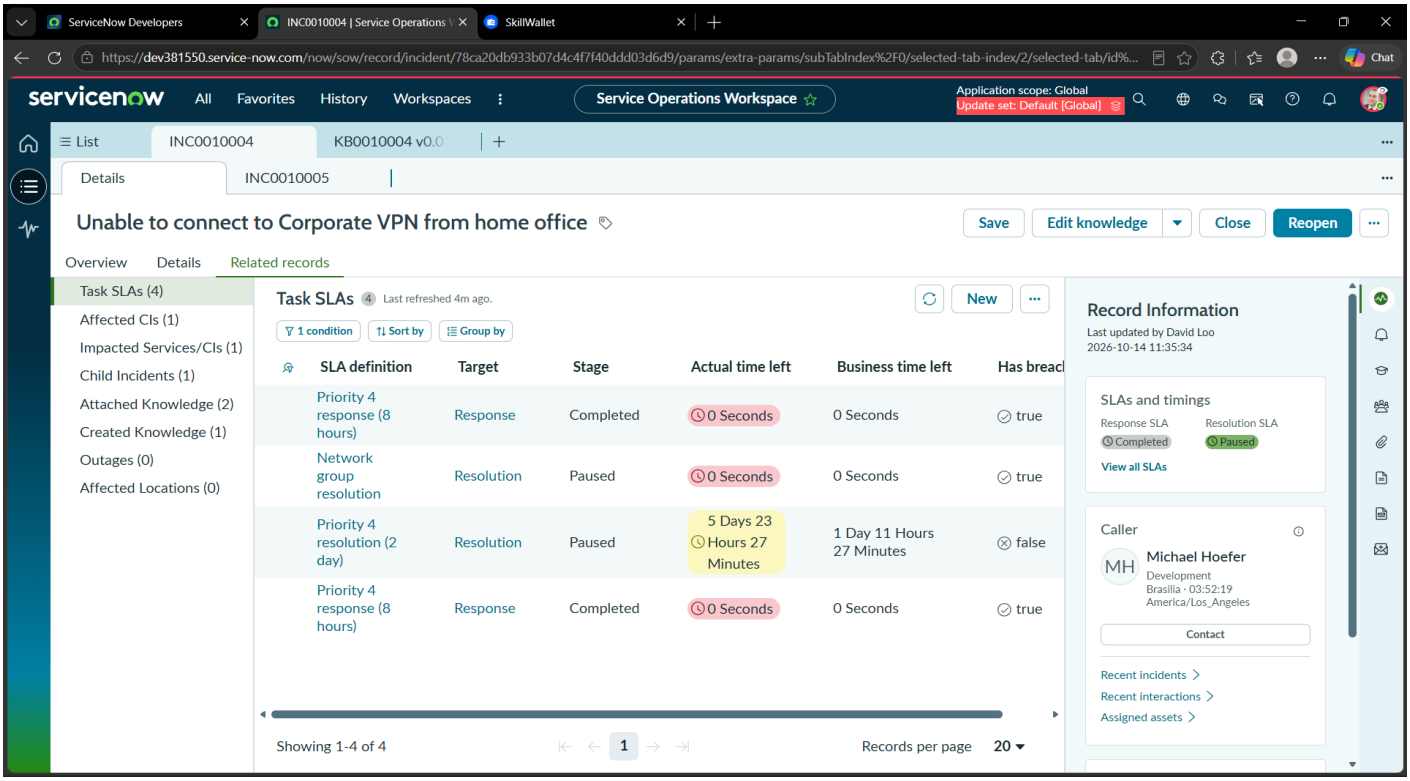


Figure 18. Final validation of the completed incident lifecycle showing affected CI, child incident, SLA records, attached knowledge, created knowledge, and Level 2 assignment.

20. End-to-End Project Flow

The completed implementation can be summarized as follows:

CORPORATE VPN ISSUE

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▼

Incident INC0010004

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▼

Classification

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▼

Knowledge Assistance

|

▼

Service Desk → Network

|

▼

Level 2: David Loo

|

▼

Configuration Item

PowerEdge T110 II

|

▼

Awaiting Change

|

▼

Child Incident Created

INC0010005

|

▼

Incident Resolution

|

▼

Resolved

|

▼

Knowledge Creation

|

▼

Final Validation

21. Implementation Summary

The project demonstrated the following ServiceNow capabilities:

Incident Management

Creation, classification, assignment, escalation, and resolution of incidents.

Service Management

Association of incidents with:

- Remote Access service
- Corporate VPN service offering

Configuration Management

Association of the incident with the affected CI:

PowerEdge T110 II

Knowledge Management

Use of Agent Assist and creation of a new knowledge article from the resolved incident. **SLA**

Management

Monitoring of incident-related SLA records.

Related Incident Management

Creation and tracking of child incident INC0010005.

Service Operations Workspace

The complete incident lifecycle was demonstrated through Service Operations Workspace.

22. Project Outcomes

The project achieved the following outcomes:

1. Standardized incident creation.
2. Structured incident classification.
3. Service and CI association.
4. Knowledge-assisted troubleshooting.
5. Support-group reassignment.
6. Level 2 incident investigation.
7. SLA visibility.
8. Child incident management.
9. Cause and resolution documentation.
10. Knowledge article generation.
11. End-to-end incident lifecycle demonstration.

23. Testing and Validation Summary

Although a separate formal testing phase was not performed, the implemented workflow was validated during execution by verifying the expected records and states.

Functionality	Validation Result
Service creation	Completed
Service offering creation	Completed
Incident creation	Completed
Incident classification	Completed
Knowledge integration	Completed
Incident reassignment	Completed
Level 2 investigation	Completed
SLA visibility	Verified
CI association	Verified
Child incident creation	Completed
Child incident resolution	Verified
Parent incident resolution	Completed
Knowledge creation	Completed
Related records validation	Completed

Note: The project execution did not include creation of a separate Change Request; the incident was placed in **On Hold – Awaiting Change** as specified in the provided implementation instructions.

24. Challenges and Observations

During implementation, several aspects of ServiceNow's Incident Management process were observed.

1. Service and Service Offering Relationships

The incident required the correct service and service offering to establish proper service context.

2. Assignment Group Behavior

Changing the assignment group from Service Desk to Network automatically cleared the previous assignee, allowing a Network technician to take ownership.

3. Related Records

ServiceNow provides multiple related lists that allow incidents to be connected with:

- SLAs
- CIs
- Knowledge
- Child incidents

- Services

4. Incident State Management

The incident lifecycle demonstrated how an incident can move between different operational states such as:

New → In Progress → On Hold → Resolved 5. Knowledge Reusability

The ability to create knowledge from resolved incidents provides an effective way to convert troubleshooting experience into reusable organizational knowledge.

25. Conclusion

The **Incident Lifecycle Automation in ServiceNow** project demonstrates a structured approach to managing IT incidents from initial reporting through resolution and knowledge creation.

The project successfully demonstrates incident creation, classification, service and configuration item association, knowledge integration, support-group reassignment, Level 2 investigation, SLA tracking, child incident management, resolution documentation, and knowledge article creation.

The integration of these ServiceNow capabilities provides improved visibility and traceability throughout the incident lifecycle.

The project also demonstrates how ServiceNow can help organizations standardize IT support processes and maintain reusable knowledge for future incidents.

Overall, the implementation provides a practical demonstration of an end-to-end Incident Management lifecycle using ServiceNow.

26. Final Project Evidence

The following screenshots provide visual evidence of the completed implementation:

Figure	**Evidence**
Figure 1	Remote Access Service
Figure 2	Corporate VPN Service Offering
Figure 3	Incident Creation
Figure 4	Incident Classification
Figure 5	Agent Assist
Figure 6	Attached Knowledge
Figure 7	Watch List / Work Notes
Figure 8	Network Reassignment
Figure 9	Level 2 Investigation
Figure 10	SLA Information
Figure 11	Awaiting Change

Figure 12	Child Incident Creation
Figure 13	Parent–Child Relationship
Figure 14	Parent Resolution
Figure 15	Child Resolution
Figure 16	Knowledge Creation
Figure 17	Created Knowledge
Figure 18	Final Validation